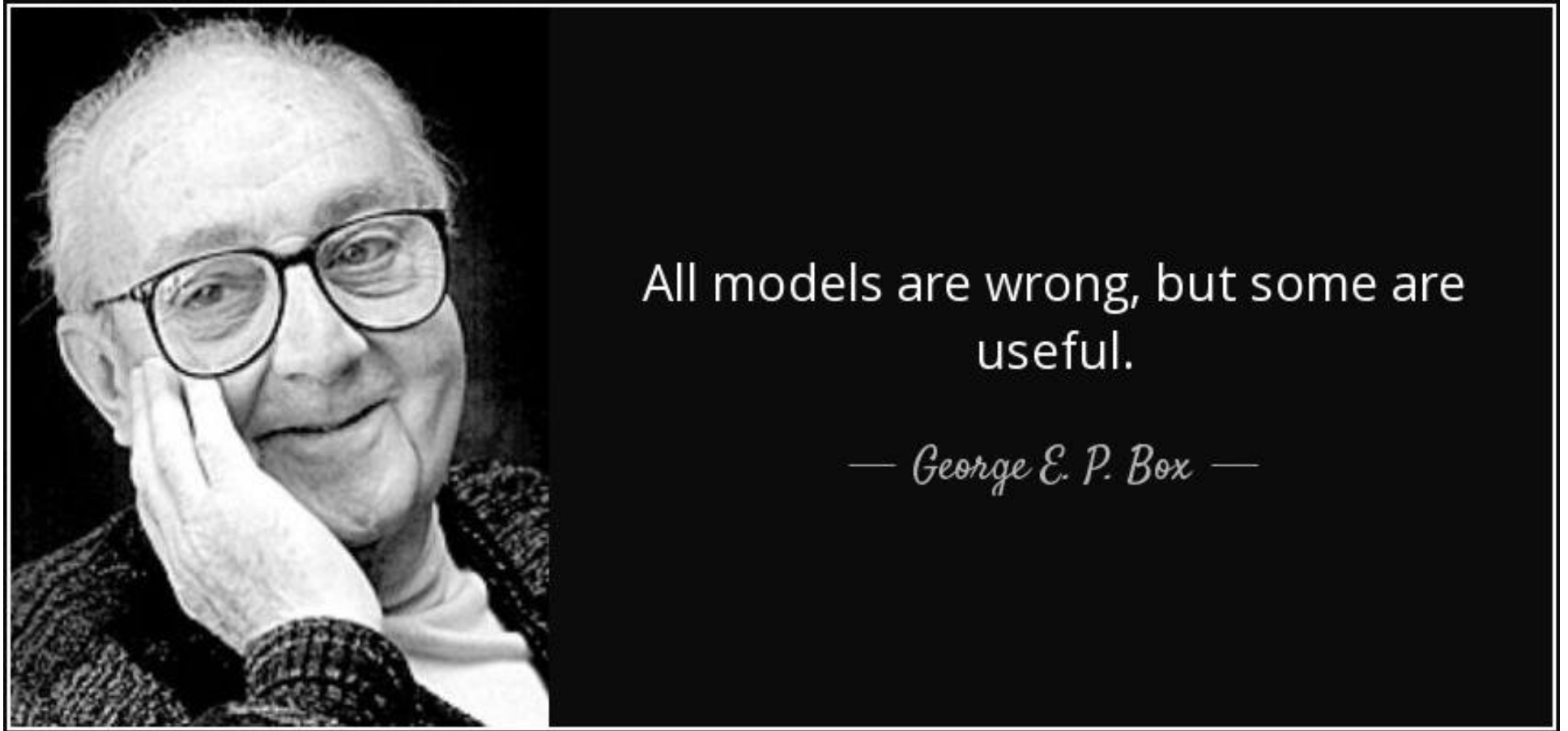


Model Checking

Announcements

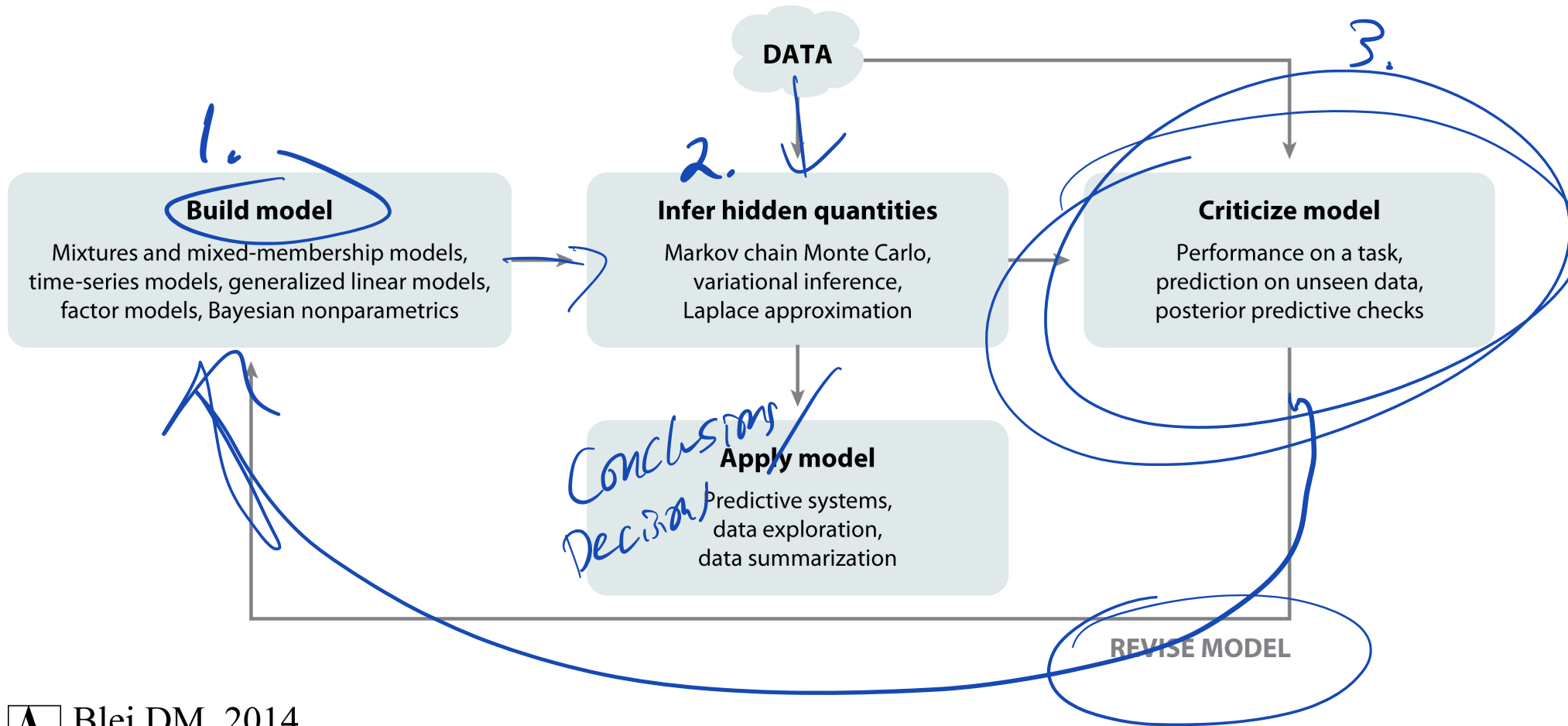
- Chapter 6 and 7 today
- Final Project Form
- HW 5 out
 - Gibbs
 - Star parameterizations
- Final Exam Studying.

All models are wrong



If the model is “wrong”, how can we improve it?

Box's Loop



Blei DM. 2014.
Annu. Rev. Stat. Appl. 1:203–32

Posterior predictive model checking

- Let y_{obs} represent the observe data $y_1, \dots, y_n$
- Let $\tilde{y}$ represent n replicated (e.g fake) observations generated from the model
- $p(\tilde{y} \mid y_{\text{obs}}) = \int p(\tilde{y} \mid \theta) p(\theta \mid y_{\text{obs}}) d\theta$
- Generate test quantity from $t(\tilde{y})$ (summary statistic)
- Check if the simulated test quantities are similar to the observed test quantity, $t(y_{\text{obs}})$

*same says
as actual
data.*

*This can be done
with Monte Carlo.*

Posterior predictive model checking

- If the model fits the data, then fake data generated under the model should look similar to the observed data
- Discrepancies can be due to model misfit or chance (or both!)
- Monte Carlo approach:
 1. sample $\theta^{(s)} \sim p(\theta | Y = y_{\text{obs}})$
 2. sample $\tilde{y}^{(s)} = (\tilde{y}_1^{(s)}, \dots, \tilde{y}_n^{(s)}) \sim \text{i.i.d. } p(y | \theta^{(s)})$
 - $\tilde{y}$ has same number of observations as y_{obs}
 3. compute $t^{(s)} = t(\tilde{y}^{(s)})$

$\theta \sim$

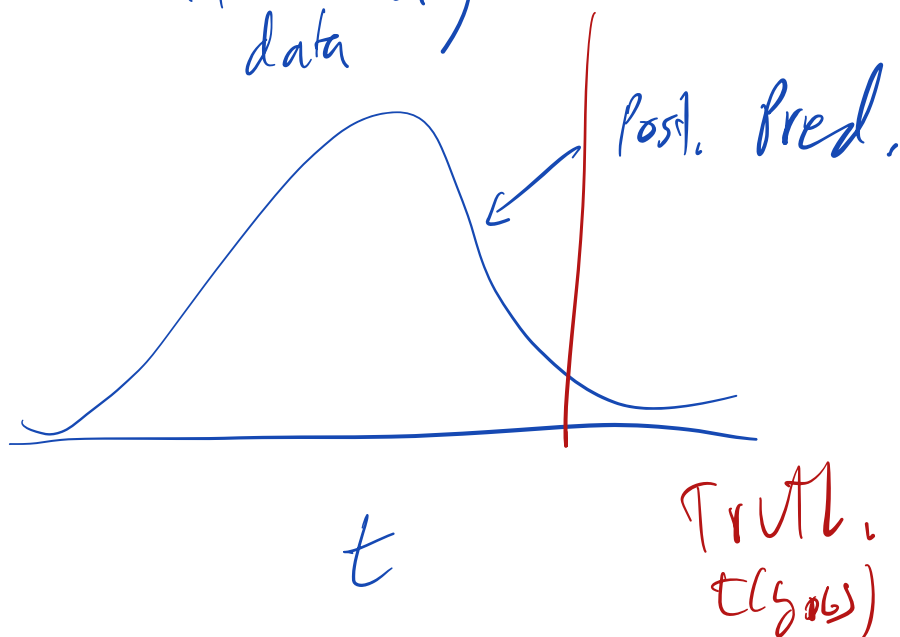
obs

"posterior
predicted
distribution
of t "

Compare t_{obs} to $\tilde{t}_{rep.}$

One stat
(from real)
data

Distribution



$(y) | y$

A Bayesian Modeling Process (overview)

1. Propose a sampling model and prior distributions
2. Compute the posterior distribution, here
$$p(\theta \mid Y = y) \sim \text{Gamma}(a + y, \beta + \nu)$$
3. Simulate test statistics, $T(\tilde{y})^{(s)}$ from the posterior predictive distribution
 - for s in $1:\text{num_fake_data}$
 - Sample $\theta^{(s)} \sim \text{Gamma}(a + y, b + \nu)$
 - Sample $\tilde{y}^{(s)} \sim \text{i.i.d Pois}(\theta^{(s)})$ (same sample size as y_{obs})
 - Compute $T(\tilde{y}^{(s)})$

4. Compare the samples $T(\tilde{y}^{(s)})$ to $T(y_{\text{obs}})$
5. Identify any model misfit, go back to step 1 and repeat.

Choosing Test Statistics

- Test quantity, or discrepancy measure, $T(y, \theta)$ can also be a function of parameters
- A model can be inadequate for some purposes but adequate for others
- Avoid choosing sufficient statistics
- Statistics should be relevant to the scientific purpose
- Often measures a feature of the data not directly addressed by the probability model (ancillary statistics)
 - e.g. Ranks of the sample or correlation of residuals

Speed of Light Example

- Newcomb measured the amount of time required for light to travel a distance of 7442 meters.
- 66 measurements were collected

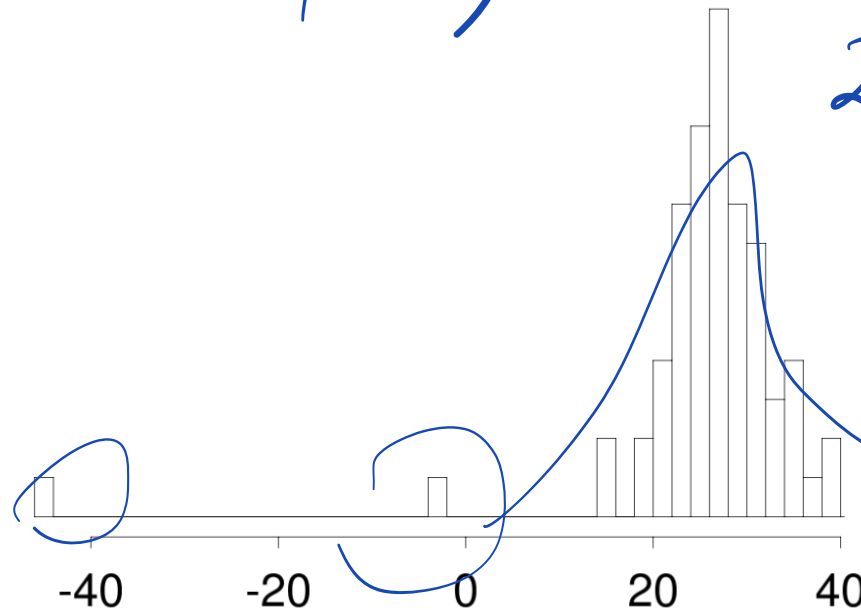
$$y_1, \dots, y_{66} \sim N(\mu, \sigma^2)$$

$$P(\mu, \sigma^2 | y_1, \dots, y_{66})$$

1. $\mu^s, \sigma^s \sim \text{Posterior}$

2. Generate
 $\tilde{y}_1, \dots, \tilde{y}_{66} \sim N(\mu^s, \sigma^s)$

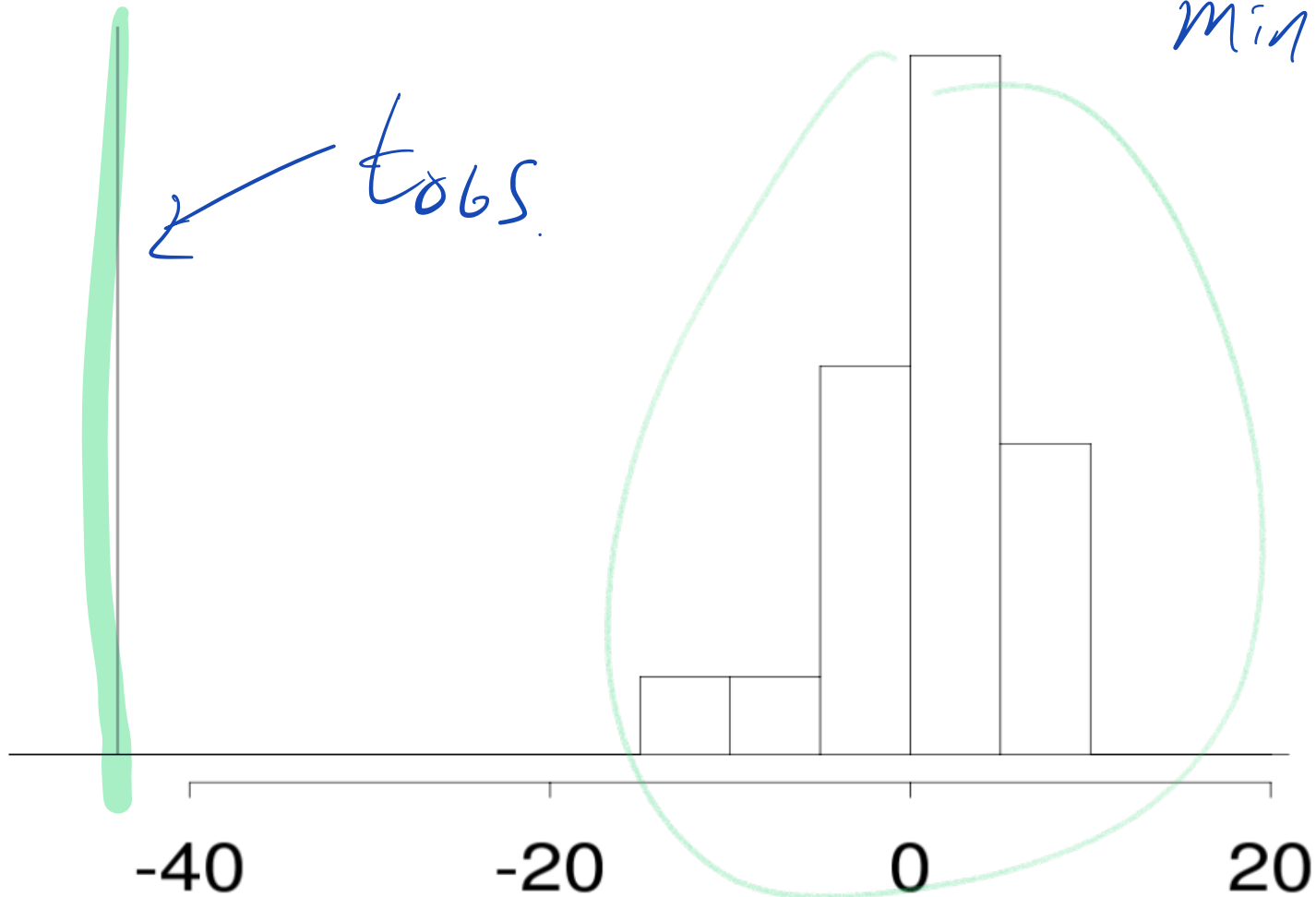
3. Compute
 $t(\tilde{y}_1, \dots, \tilde{y}_{66})$



Speed of Light ($T = \text{min}$)

$$t(\tilde{y}_1, \dots, \tilde{y}_{66}) =$$

$$\min(\tilde{y}_1^2, \dots, \tilde{y}_{66})$$

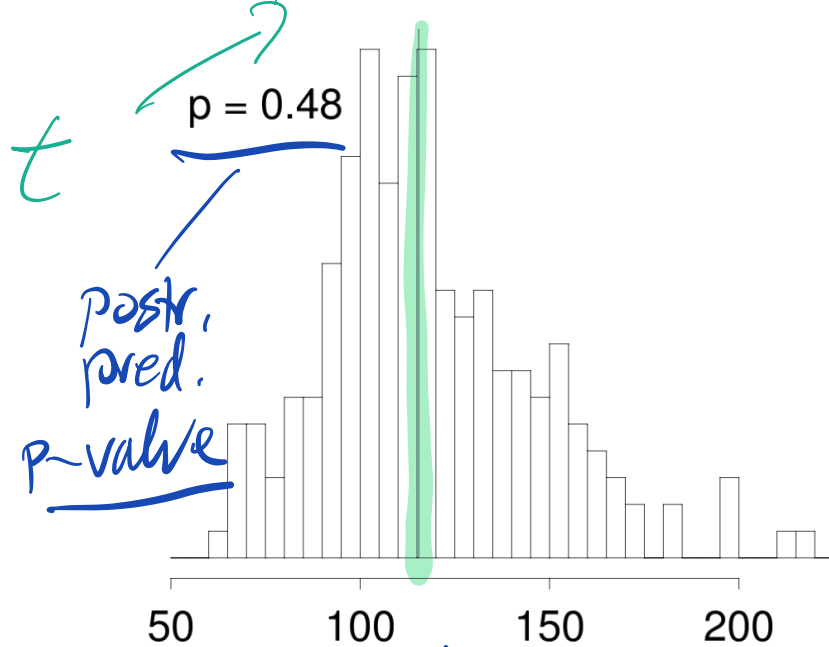


$$\hookrightarrow P(\min(\tilde{y}_1, \dots, \tilde{y}_{66}) | y_{obs})^{10}$$

Speed of Light Example

$$T(y, \theta) =$$

$$P(\text{var}(\tilde{y}_1, \dots, \tilde{y}_{66}) | y_{\text{obs}})$$



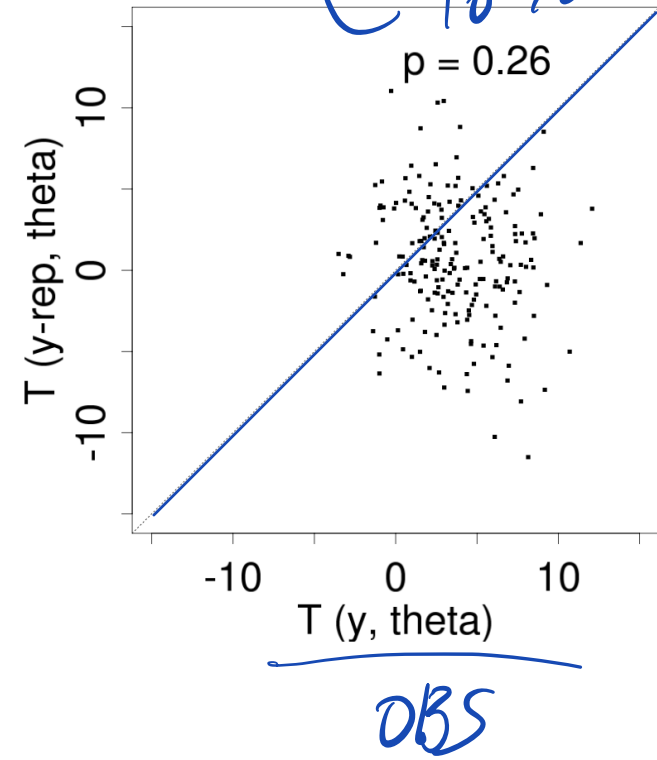
Good fit
for variance.

$$\frac{|\tilde{y}_{(b)} - \mu|}{\sqrt{\text{var}(\tilde{y}_{(b)})}}$$

90% quantile

10%

Rep.



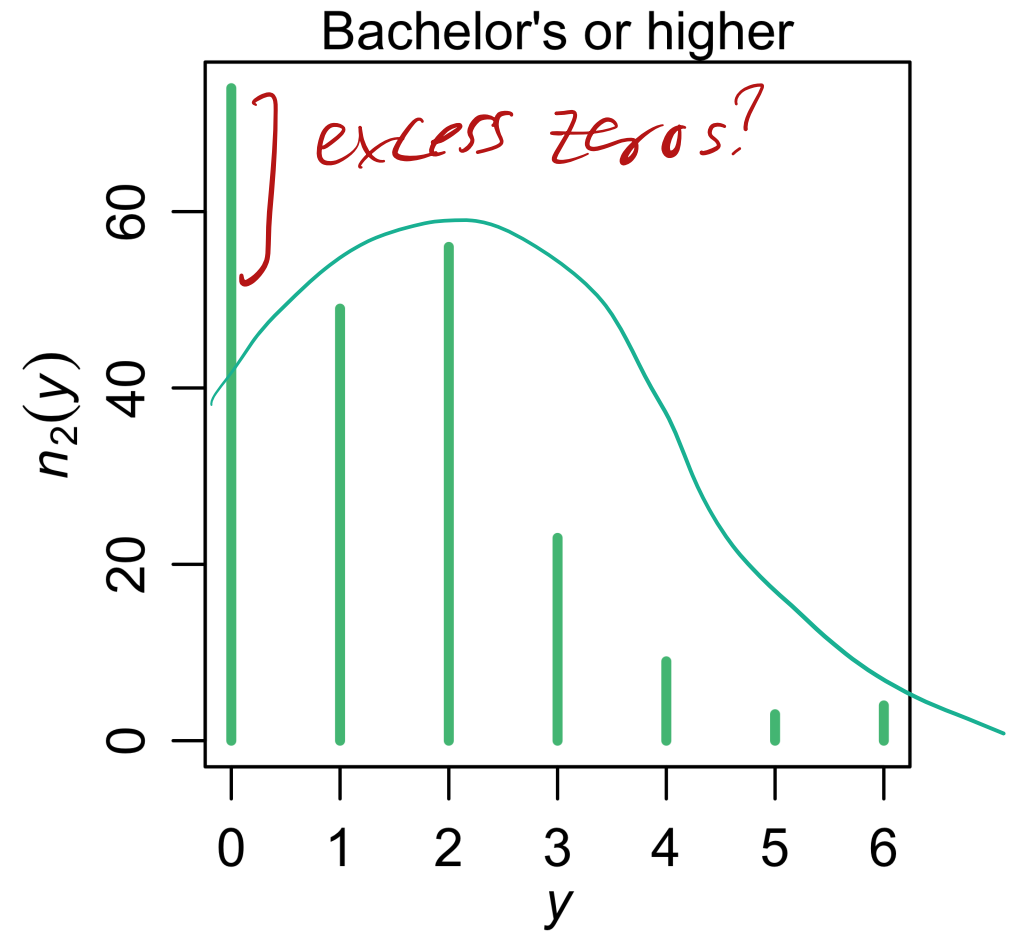
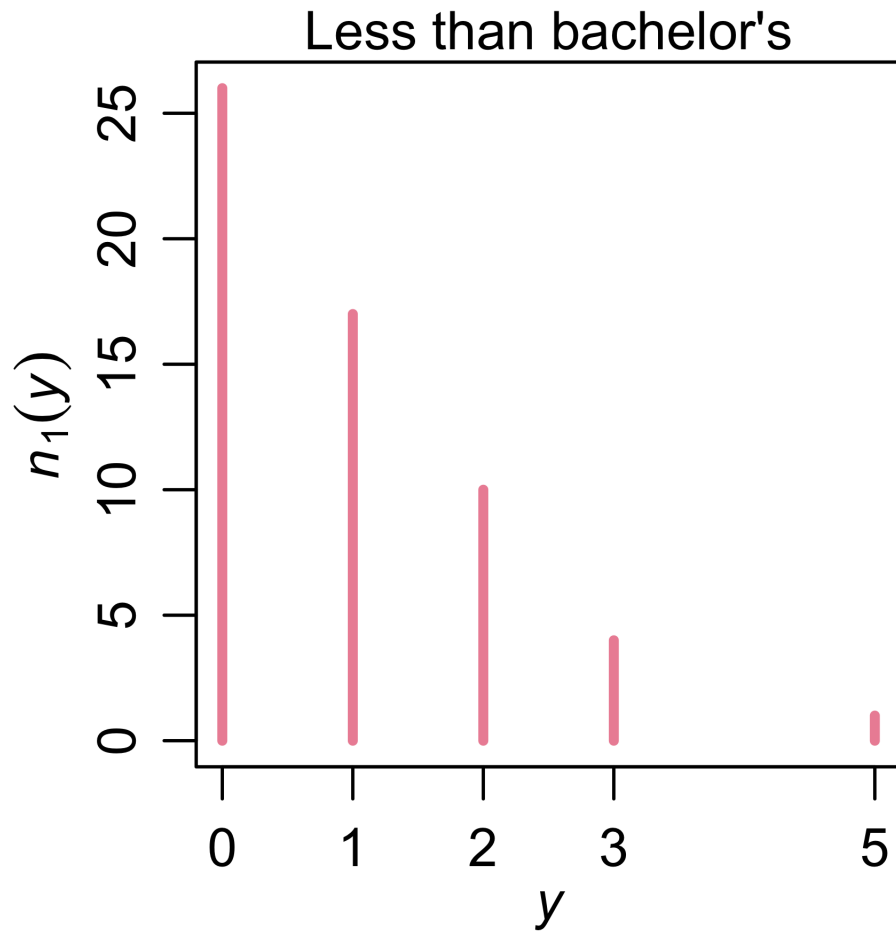
Predictive Checks: an example

- In the 1990's there was a survey of 155 women, at least 40 years of age
- Recorded number of children and educational attainment
 - Bachelor's degree or higher ($n_1 = 111$)
 - Less than bachelor's degree ($n_2 = 44$)

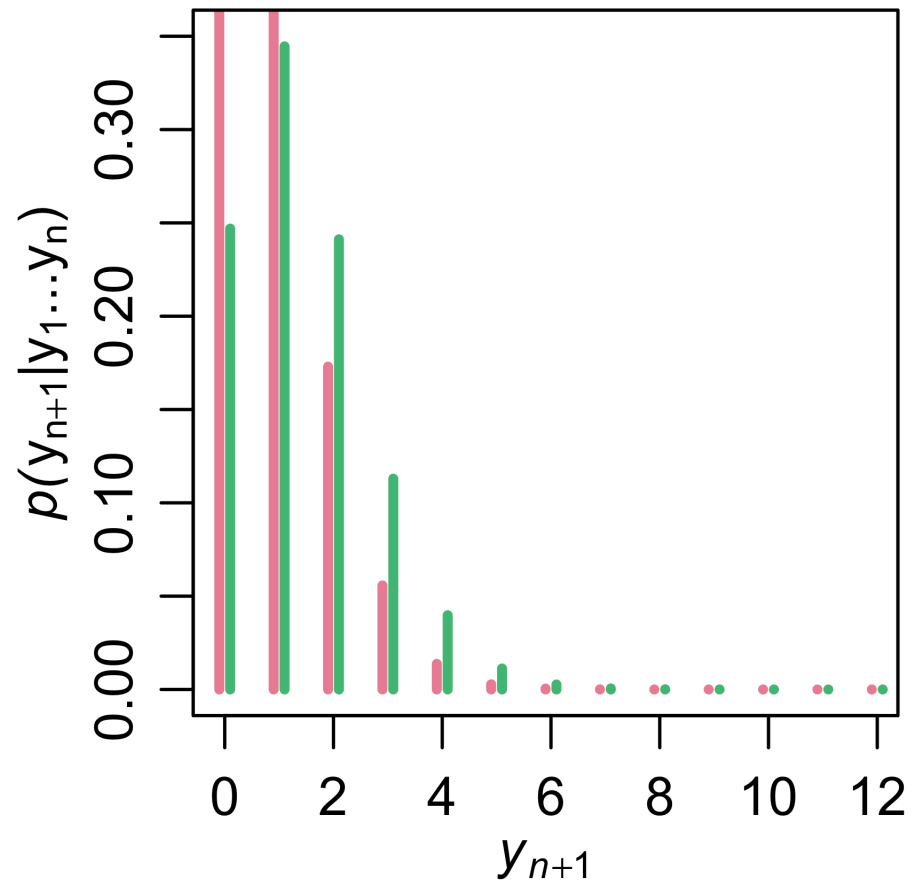
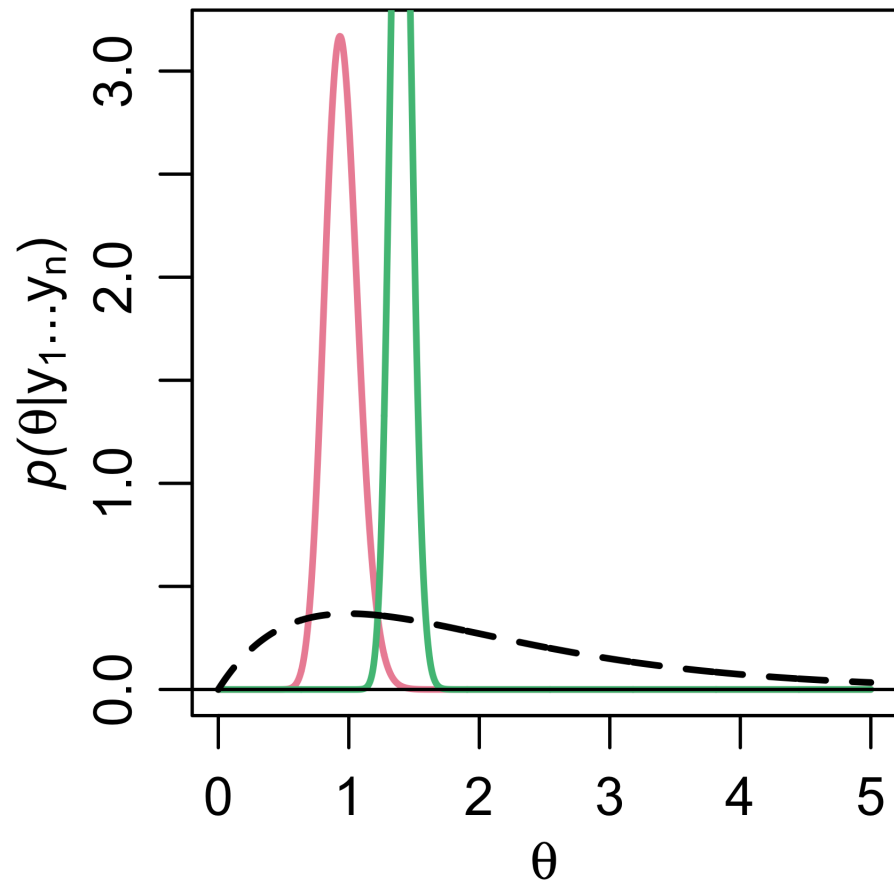
$$Y_{1,1} \dots, Y_{n_1,1} | \theta_1 \sim \text{i.i.d. Poisson}(\theta_1)$$

$$Y_{1,2} \dots, Y_{n_2,2} | \theta_2 \sim \text{i.i.d. Poisson}(\theta_2)$$

PPCs example



PPCs example



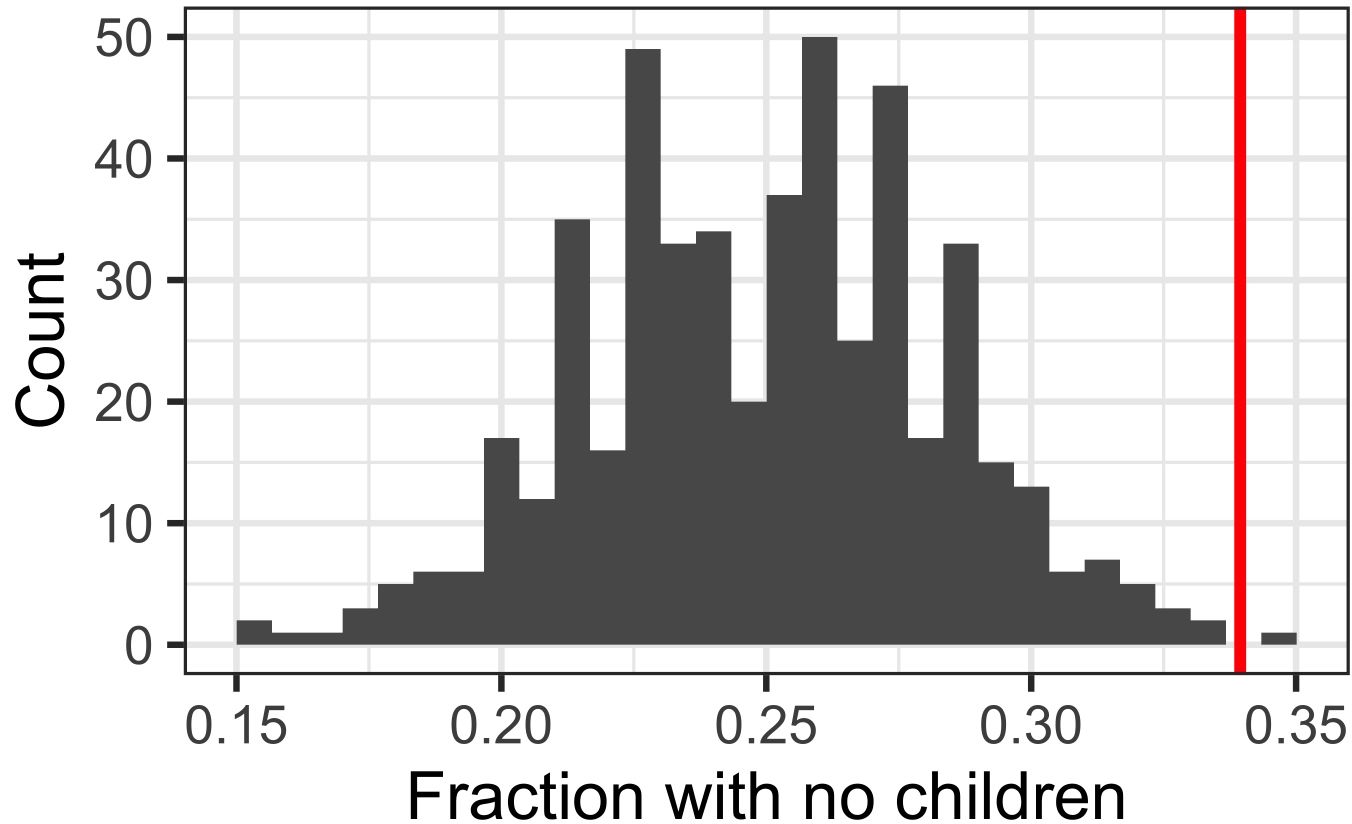
PPCs example

- Let's check the model fit for the “without Bachelor's” group first
- Do S times:
 - sample $n_2 = 44$ observations $\tilde{y}$ from the posterior predictive distribution
- Let $T(\tilde{y})$ be the fraction of women with no children

```
1  S <- 1000
2  t_s <- numeric(S)
3  for(s in 1:S){
4    theta_s <- rgamma(1, a, b) # whatever a and b are for my posterior
5    ytilde_s <- rpois(n=44, theta = theta_s)
6    t_s[s] <- mean(ytilde_s == 0) # compute test stat
7  }
8
9  ## then visualize histogram of t_s
```

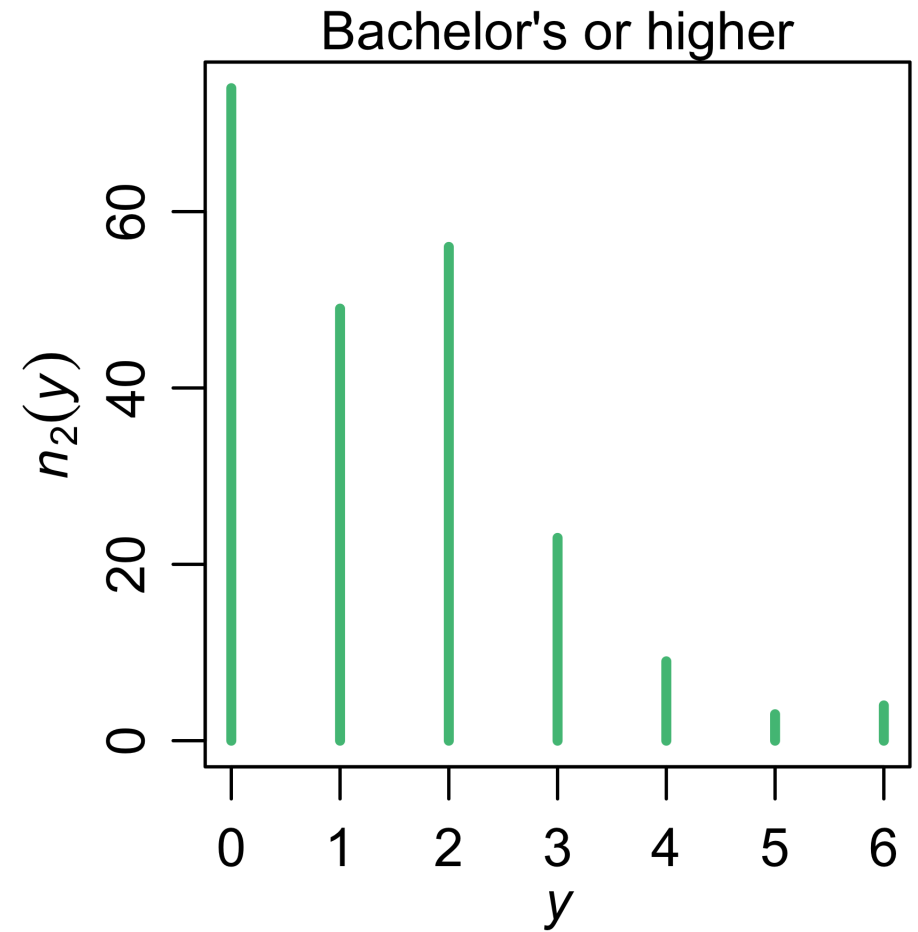
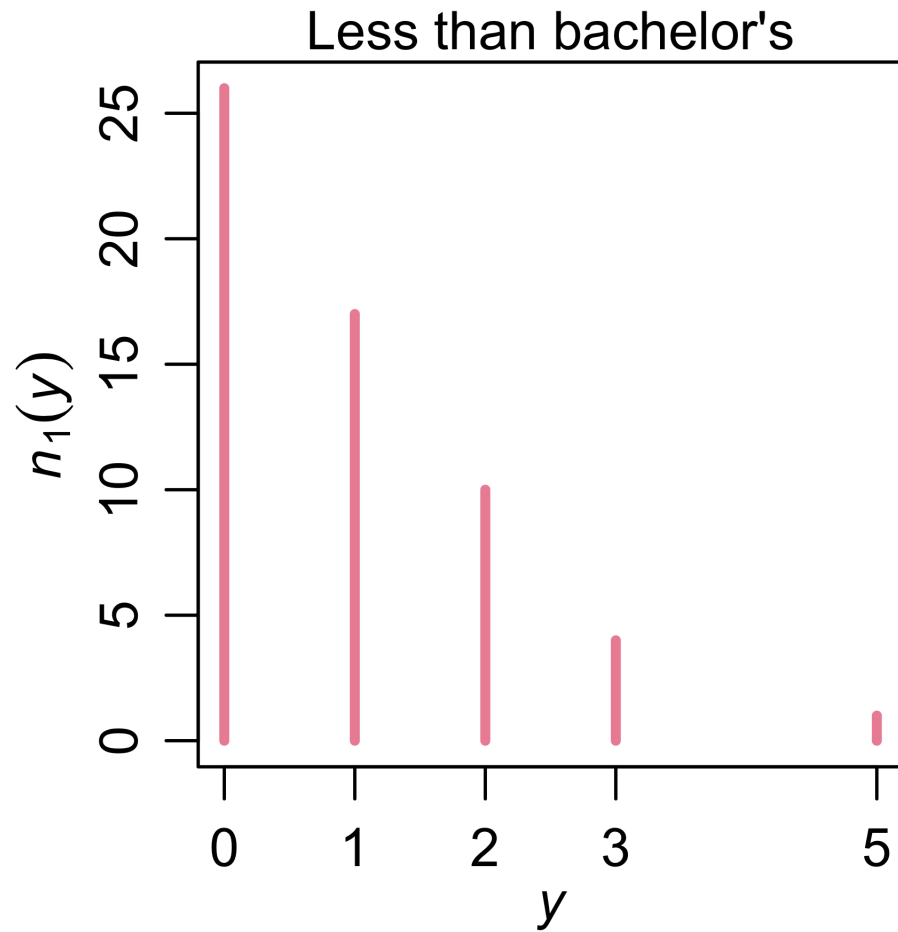
PPCs example

Women without Bachelors degree



$$Pr((T^{rep} > T^{obs} \mid y_{obs}) = 0.002$$

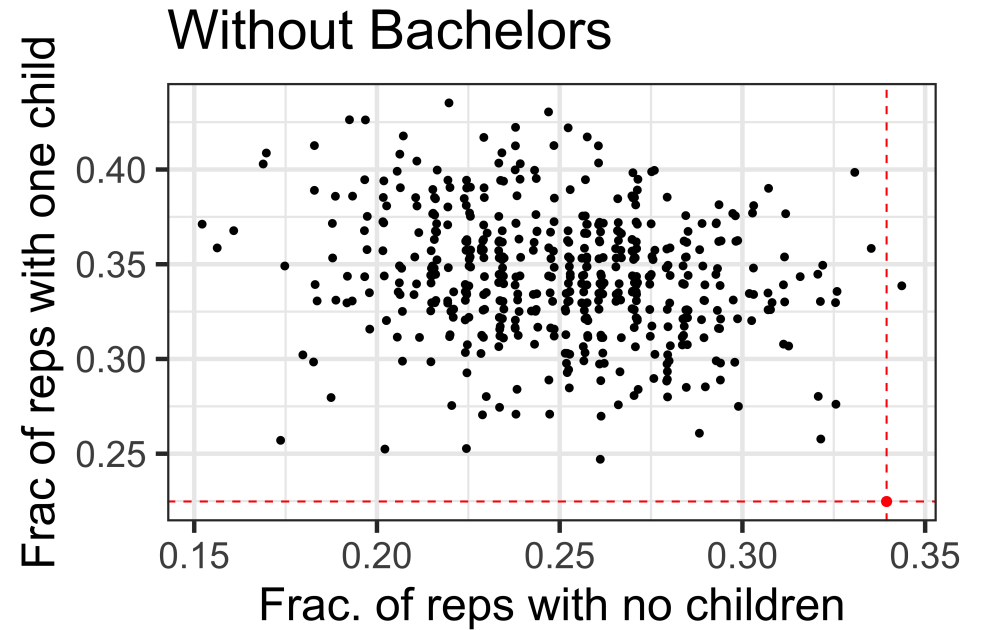
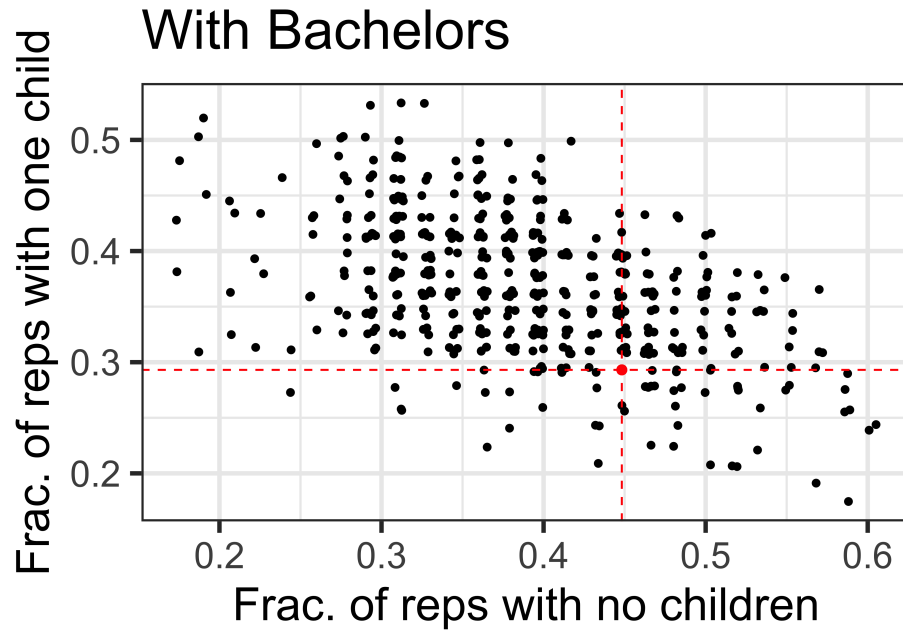
PPCs example



PPCs example

- Model checking both groups
- Look at fit for two different test statistics:
 - Fraction with no children
 - Fraction with one child

Poisson example



Predictive Distributions in Stan

```
1  ...
2  } model {
3    y1 ~ poisson(lambda1);
4    y2 ~ poisson(lambda2);
5  } generated quantities{
6
7    int y1rep[n1];
8    int y2rep[n2];
9    real log_lik[n1+n2];
10   for(i in 1:n1){
11     y1rep[i] = poisson_rng(lambda1);
12     log_lik[i] = poisson_lpmf(y1[i] | lambda1);
13   }
14   for(i in 1:n2){
15     y2rep[i] = poisson_rng(lambda2);
16     log_lik[n1+i] = poisson_lpmf(y2[i] | lambda2);
17   }
18 }
```

Predictive Checks in Stan

Running MCMC with 4 sequential chains...

Chain 1 finished in 2.0 seconds.

Chain 2 finished in 2.1 seconds.

Chain 3 finished in 2.3 seconds.

Chain 4 finished in 2.0 seconds.

All 4 chains finished successfully.

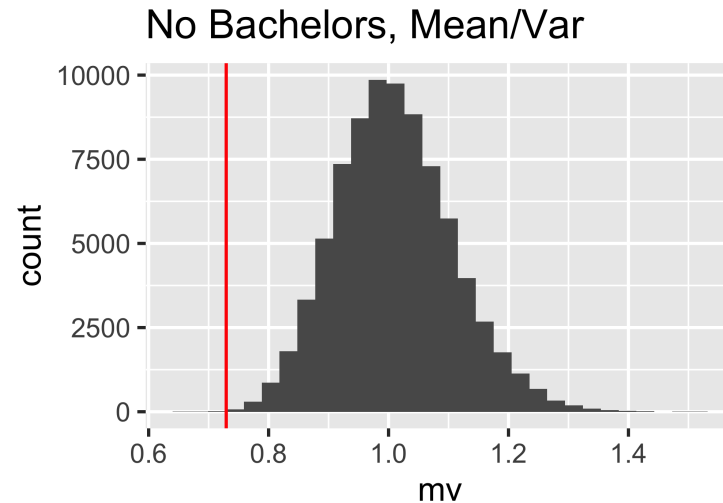
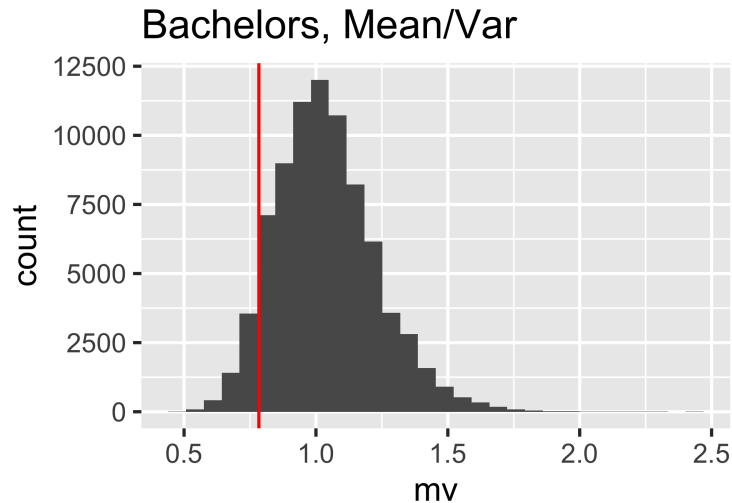
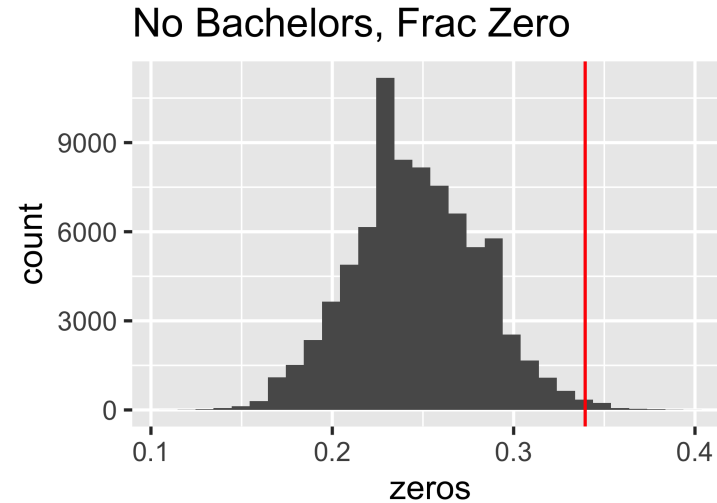
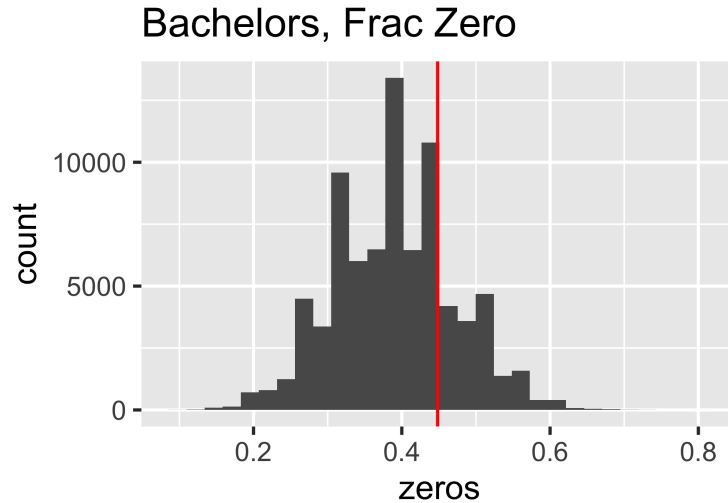
Mean chain execution time: 2.1 seconds.

Total execution time: 8.5 seconds.

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp__	-261.16	-260.85	1.02	0.71	-263.21	-260.21	1.00	35823	47636
lambda1	0.95	0.94	0.13	0.13	0.75	1.17	1.00	75976	54495
lambda2	1.40	1.40	0.08	0.08	1.27	1.54	1.00	75256	53473
y1rep[1]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	80952	80389
y1rep[2]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	79270	79427
y1rep[3]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	79299	78493
y1rep[4]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	78656	80525
y1rep[5]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	78714	78498
y1rep[6]	0.96	1.00	0.99	1.48	0.00	3.00	1.00	80345	79532
y1rep[7]	0.94	1.00	0.98	1.48	0.00	3.00	1.00	79840	79077



Predictive Checks (Pois. Model)



ZIP (mixture model)

$$Z \sim \text{Bern}(\phi)$$

$$Y = 0 \text{ if } Z = 1$$

$Z = 1$ means "Can't have children"

$$Y \sim \text{Pois}(\lambda) \text{ if } Z = 0$$

$$P(Y|\lambda) = \phi \mathbb{I}[Y=0] + (1-\phi) e^{-\lambda} \frac{\lambda^Y}{Y!}$$

ZIP

Running MCMC with 4 sequential chains...

Chain 1 finished in 0.4 seconds.

Chain 2 finished in 0.4 seconds.

Chain 3 finished in 0.5 seconds.

Chain 4 finished in 0.5 seconds.

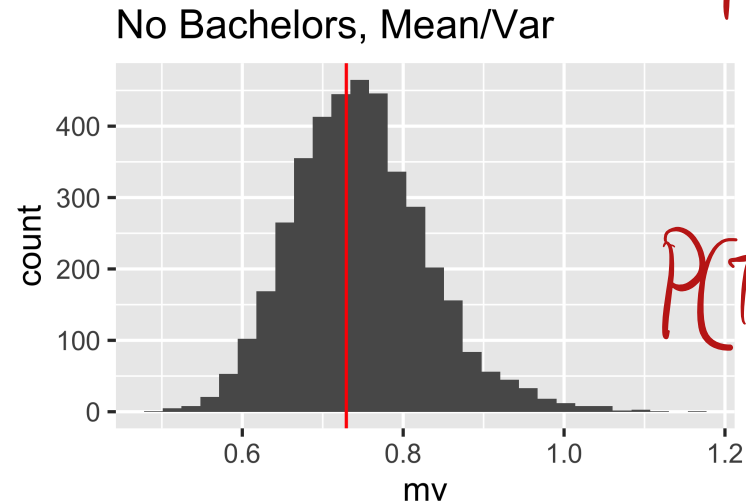
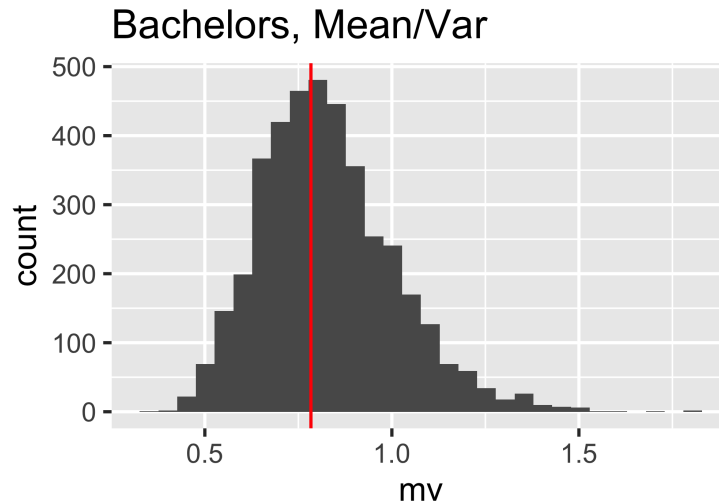
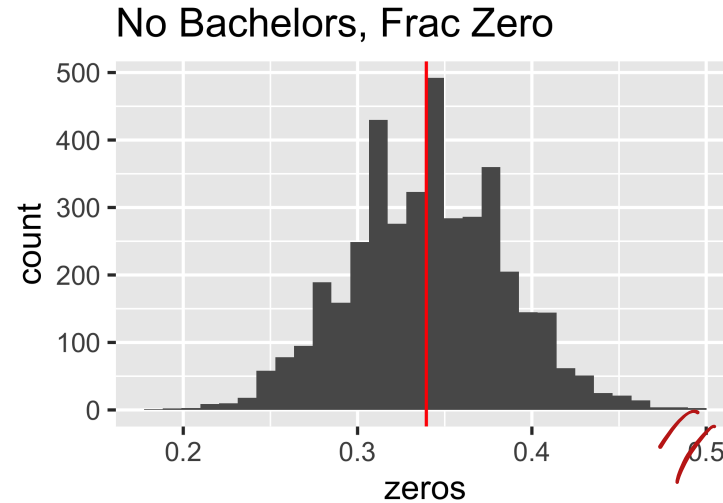
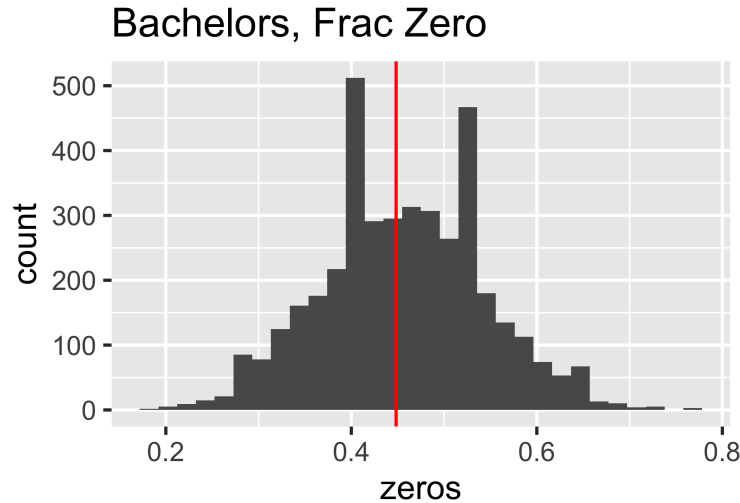
All 4 chains finished successfully.

Mean chain execution time: 0.5 seconds.

Total execution time: 2.0 seconds.

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp__	-423.61	-423.24	1.54	1.28	-426.63	-421.84	1.00	1454	2020
lambda1	1.20	1.18	0.21	0.21	0.89	1.58	1.00	2372	2583
lambda2	1.75	1.75	0.13	0.13	1.54	1.96	1.00	2604	2453
theta1	0.21	0.21	0.10	0.11	0.04	0.38	1.00	2053	1306
theta2	0.20	0.20	0.04	0.04	0.13	0.27	1.00	2503	1935
ylrep[1]	0.94	1.00	1.12	1.48	0.00	3.00	1.00	3939	3904
ylrep[2]	0.96	1.00	1.11	1.48	0.00	3.00	1.00	3790	3780
ylrep[3]	0.94	1.00	1.10	1.48	0.00	3.00	1.00	3330	3069
ylrep[4]	0.92	1.00	1.09	1.48	0.00	3.00	1.00	3215	3372
ylrep[5]	0.94	1.00	1.10	1.48	0.00	3.00	1.00	3984	3862

Predictive Checks (ZIP Model)



Posterior // Predictive P-value
 $P(T_{obs} > T_{rep} | y)$

Not a true p-value!

Model Comparison

- PPCs for checking model fit. But how do we compare different models (regardless of fit)?
- Compare models by their out-of-sample predictive power
- Leave-one-out cross validation
 - Fit model n times, leaving out one observation
 - Evaluate predictive accuracy on left out observations
 - Typically used for parameter tuning
- Can use ideas from LOO-CV to compare models

Measures of Predictive Accuracy

- Point prediction: predict a single unknown future observation *held out*
- Measures of predictive accuracy are called scoring functions (e.g. MSE)
- Probabilistic predictions account for uncertainty
 - **Scoring rules:** measure of accuracy based on probabilistic predictions
 - Examples: quadratic, logarithmic and zero-one scores

Log-likelihood as a scoring rule

$$y \sim \mathcal{N}(\mu, \sigma^2)$$
$$\text{ld.} \quad -\frac{(y - \mu)^2}{2\sigma^2}$$

- As a measure of accuracy, we call it the “log predictive density”
- $\log(p(y|\theta))$ is proportional to MSE if the data are normal with constant variance
- Connection to Kullback-Leibler divergence
 - KL: $-E_p \log(q(y)/p(y)) = -E_p \log(q(y)) + E_p \log(p(y))$
 - Asymptotically, the lowest KL model is the one with the highest expected log predictive density

Measure of "distance" between distributions.

Chosen model

True Model.

Make as big as possible.

Out-of-sample prediction

Ideal: Predict on out-of-sample data

$$\log P_{\text{p.p.}}(\tilde{y}) = \log E_{P(\theta/y)} P(\tilde{y}|\theta) =$$

$$\log \int P(\tilde{y}|\theta) P(\theta/y) d\theta$$

$\tilde{y}$ comes from true data generating model, not $P(\tilde{y}|\theta)$

Expected Log Pointwise Predictive Density

ELPD (one obs)

$$E_f \log_{p_f} p(\tilde{y}) = \int \log_{p_f} p(\tilde{y}) f(\tilde{y}) d\tilde{y}$$

$f(\tilde{y})$ is true
data generating model.

Average the
log predictive
density (for
candidate model)
over all $f(\tilde{y})$ with

Problem: Don't know the true model, f ,

Soln. $\widehat{ELPD}_{LOO} : \frac{1}{n} \sum_{i=1}^n \log P(y_i / y_{-i})$

$\uparrow$
approximates
 E_f

log posterior
predictive,
holding out
 i , evaluating on
 i .

Problem 2:

Need

$$P(\theta / y_{-i}) \quad \forall i$$

MCMC is SLOW

I have: $P(\theta / y)$
(posterior given
full data)

$$\int P(y_i / \theta) P(\theta / y_{-i}) d\theta$$

Solution: Importance Sampling.

$$\int P(y_i | \theta) P(\theta | y_{-i}) d\theta =$$

$$\int P(y_i | \theta) \frac{P(\theta | y_{-i}) P(\theta | y)}{P(\theta | y)} d\theta$$

Use samples
from full
posterior

weight

to approximate
leave-1-out posterior. (close to 1?)

$$P(y_i | y_{-i}) = \int w_i(\theta) P(y_i | \theta) P(\theta | y) d\theta$$

When $y_1, \dots, y_n$ are iid:

$$w_i(\theta) = \frac{P(\theta / y_{-i})}{P(\theta / y)} = \frac{1}{P(y_i / \theta)}$$

$$P(\theta / y) \propto P(\theta) \prod_{i=1}^n P(y_i / \theta)$$

$$P(\theta / y_{-i}) \propto P(\theta) \prod_{j \neq i} P(y_j / \theta)$$

Pareto-Smoothed Importance Sampling

+ could see (truth)

- When the data are conditionally independent,

$$r_i^s = \frac{1}{p(y_i|\theta^s)} \propto \frac{p(\theta^s|y_{-i})}{p(\theta^s|y)} \text{ to get importance sampling estimates}$$

PSIS-LOO

- $p(\tilde{y}_i | y_{-i}) \approx \frac{\sum_{s=1}^S r_i^s p(\tilde{y}_i | \theta^s)}{\sum_{s=1}^S r_i^s}$

- $p(y_i | y_{-i}) \approx \frac{1}{\frac{1}{S} \sum_{s=1}^S \frac{1}{p(y_i|\theta^s)}}$

<https://link.springer.com/article/10.1007/s11222-016-9696-4>

```

1 library(loo)
2 nb_model <- cmdstan_model("nb_model.stan")
3 nb_fit = nb_model$sample(
4     data=list(n1=num_bachelors, n2=num_no_bachelors,
5     y1=bachelors_data, y2=no_bachelors_data),
6     refresh=0)

```

Running MCMC with 4 sequential chains...

Chain 1 finished in 0.4 seconds.
 Chain 2 finished in 0.4 seconds.
 Chain 3 finished in 0.4 seconds.
 Chain 4 finished in 0.5 seconds.

All 4 chains finished successfully.
 Mean chain execution time: 0.4 seconds.
 Total execution time: 1.8 seconds.

```

1 loo_compare(list("pois"=pois_fit$loo(),
2     "zip"=zip_fit$loo(),
3     "nb"=nb_fit$loo()))

```

	elpd_diff	se_diff
zip	0.0	0.0
nb	-3.2	2.4
pois	-8.6	4.6

Galaxies Example

- Velocities in km/sec of 82 galaxies (Corona Borealis region).
- Multimodality in such surveys is evidence for voids and superclusters in the far universe.
- Statistical question: how many clusters are there in this dataset?

Model Comparison

```
1 library("loo")
2 galaxy_speeds <- log(MASS::galaxies)
3 galaxy_results <- list()
4
5 ## Run model for each of k clusters
6 mix_model <- cmdstanr::cmdstan_model("mix_model.stan")
7 for(k in 1:5) {
8   galaxy_results[[k]] <- mix_model$sample(data=list(K=k, N=length(galaxy_sp
9                                           refresh=0, show_messages=FALSE)
10 }
```

Warning: 63 of 4000 (2.0%) transitions ended with a divergence.

See <https://mc-stan.org/misc/warnings> for details.

Warning: 16 of 4000 (0.0%) transitions hit the maximum treedepth limit of 10.

See <https://mc-stan.org/misc/warnings> for details.

Model Comparison

```
1 loo_compare(galaxy_results[[1]]$loo(),  
2             galaxy_results[[2]]$loo(),  
3             galaxy_results[[3]]$loo(),  
4             galaxy_results[[4]]$loo(),  
5             galaxy_results[[5]]$loo())
```

	elpd_diff	se_diff
model3	0.0	0.0
model5	-0.4	1.4
model4	-1.0	0.7
model2	-10.7	3.7
model1	-41.1	8.7

